

Common-Corruptions Benchmark — Report

Run started: 2026-05-26 16:45 UTC. Author: Sachit Jain.

This report covers Phase 1, Axis A — the 15 ImageNet-C-style common corruptions at severity 3, evaluated on the 7 ImageNet baselines listed in `config.yaml`. The corruption sets are generated once by `scripts/generate_datasets.py --corruptions` and shared across all models; this report aggregates the per-(model, corruption) JSONs in this directory.

1. Setup

- **Eval host (where inference ran):** Tesla T4. torch 2.10.0+cu128, torchvision 0.25.0+cu128.
- **Report rebuilt on:** CPU, Windows-11-10.0.26200-SP0.
- **Attack:** 15 corruption types (`attacks/corruptions.py`), severity 3. Implementation: self-contained NumPy/SciPy/PIL (no `imagenet_c` / ImageMagick dependency). Calibrations follow Hendrycks & Dietterich (ICLR 2019); `motion_blur/snow/frost/fog` use NumPy approximations of the canonical Wand versions and may not be bit-equivalent to the published ImageNet-C numbers.
- **Dataset:** the full 1000-image clean benchmark, pre-cropped to 224×224 (torchvision eval transform) before corruption. One corrupted variant per (clean image, corruption type) = 15000 images total.
- **Global seed:** 42 (`config.yaml`). Per-(image, corruption) RNG seeded with (`seed + sha1(corruption_type) + image_index`) so a single corruption type can be regenerated without touching the others.

2. Headline (robust accuracy per (model, corruption))

model	gaus	shot	impu	defor	glass	moti	zoon	snow	frost	fog	bright	conti	elast	pixel	jpeg	mean	mean_drop
resnet50	0.487	0.457	0.358	0.447	0.088	0.129	0.393	0.664	0.618	0.384	0.721	0.700	0.771	0.594	0.660	0.498	0.283
vgg16	0.150	0.143	0.091	0.190	0.056	0.045	0.242	0.511	0.420	0.165	0.563	0.313	0.647	0.285	0.475	0.286	0.401
convnext_tiny	0.567	0.554	0.565	0.489	0.130	0.160	0.417	0.730	0.656	0.387	0.750	0.712	0.769	0.643	0.701	0.549	0.238
vit_b_16	0.636	0.637	0.515	0.522	0.258	0.216	0.463	0.730	0.677	0.543	0.743	0.717	0.773	0.686	0.695	0.587	0.206
swin_t	0.592	0.566	0.563	0.490	0.134	0.181	0.408	0.727	0.684	0.549	0.738	0.728	0.773	0.583	0.671	0.559	0.216
efficientnet_b3	0.432	0.436	0.425	0.353	0.093	0.110	0.337	0.609	0.536	0.226	0.695	0.526	0.725	0.585	0.647	0.449	0.294
clip_vit_b_16	0.403	0.400	0.368	0.396	0.147	0.141	0.316	0.533	0.511	0.355	0.579	0.543	0.596	0.481	0.488	0.417	0.204

Machine-readable copies: `accuracy_matrix.csv` (wide), `accuracy_table.csv` (long).

3. Sanity checks

- **Severity-3 mean drop in literature window [0.05, 0.25]:** observed **0.263** → **FAIL**.
- **Every corruption causes >1% drop on ≥5 of 7 models:** **PASS** — all 15 types meet the bar.
- **Coverage:** 105 / 105 (model × corruption) pairs reported. **PASS**.

4. Per-axis analysis

Across 7 of 7 models and 15 of 15 corruption types at severity 3, the mean accuracy drop was **0.263**. The most robust model on this axis was `clip_vit_b16` (mean drop **0.204**); the least robust was `vgg16` (mean drop **0.401**). The hardest corruption type (averaged across models) was `glass_blur` with mean drop **0.612**; the easiest was `elastic_transform` with mean drop **0.019**.

5. Figures

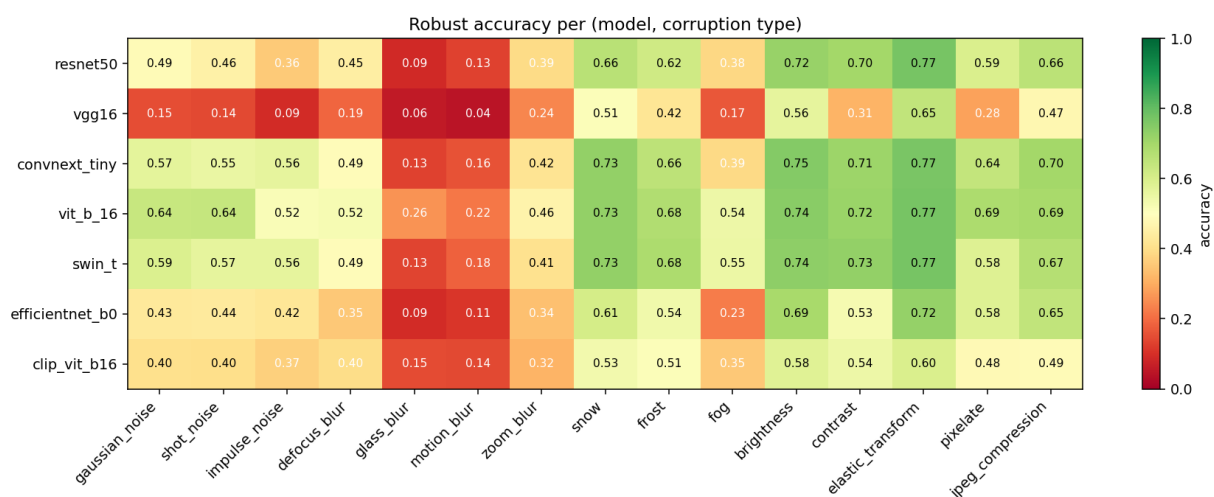


Figure 1 — Robust accuracy heatmap (7 models × 15 corruptions).

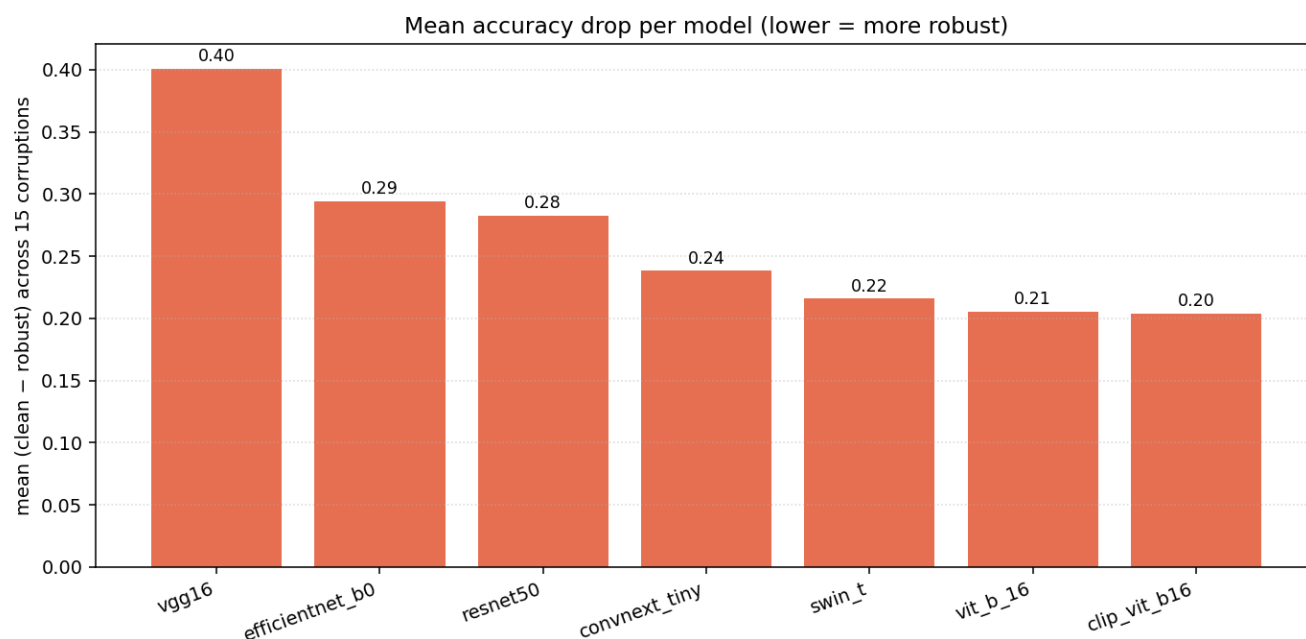


Figure 2 — Mean accuracy drop per model (averaged across 15 corruptions).

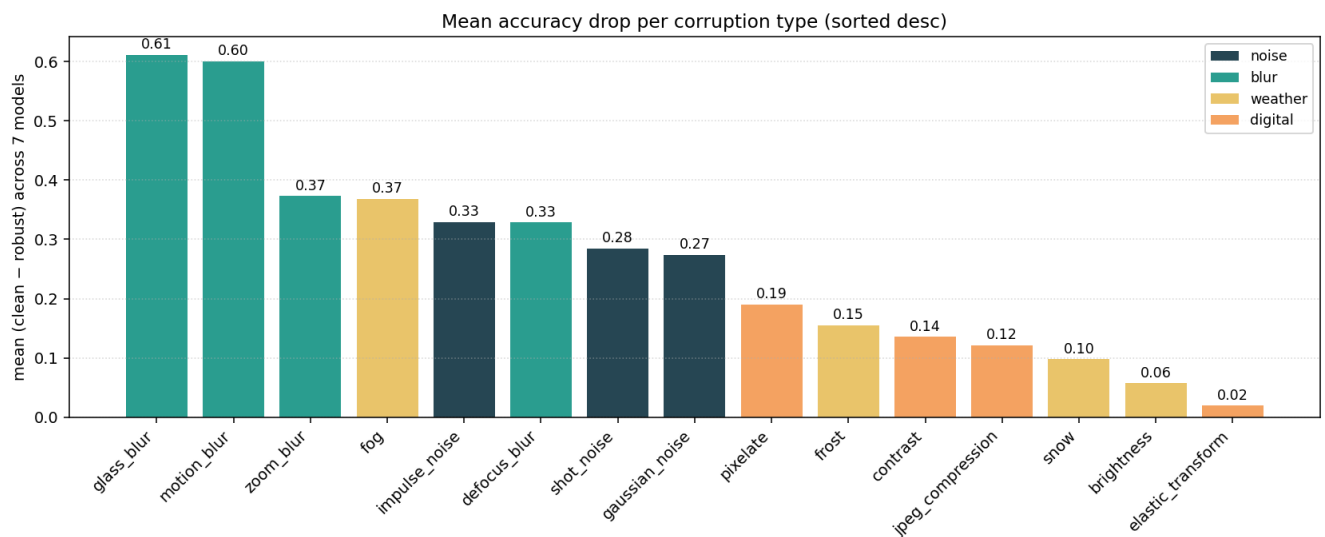


Figure 3 — Mean accuracy drop per corruption type (averaged across 7 models).

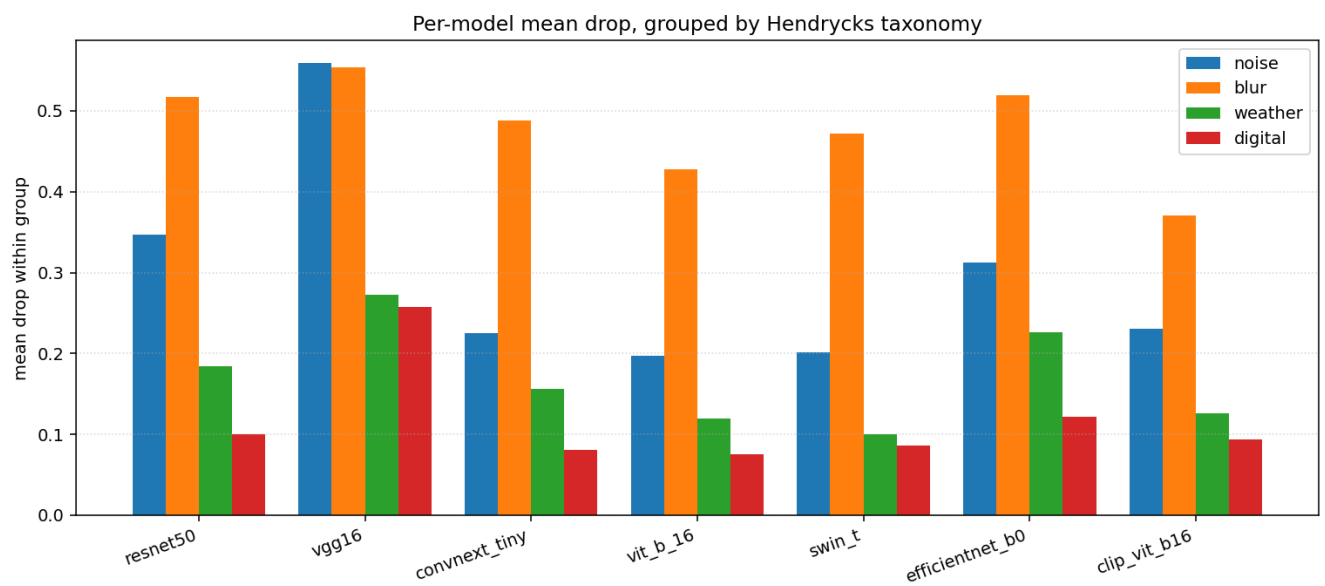


Figure 4 — Per-model mean drop grouped by Hendrycks taxonomy.



Figure 5 — One image under each corruption type.

6. Interpretation

Common corruptions are the non-adversarial, distribution-shift end of the robustness spectrum: each image is degraded by physical noise, blur, weather, or digital artifacts in a way that a human would still classify correctly. The drops here represent **realistic deployment failures** (a webcam in fog, a low-bandwidth JPEG upload, a shaky phone photo), not worst-case adversarial inputs. Unlike the gradient and typographic axes, this measure decouples natural fragility from language grounding — every model is graded on the same model-agnostic transformation. The Phase D defense will be re-evaluated on this axis to check whether adversarial fine-tuning sacrifices natural-corruption robustness as a side effect.

7. Reproducibility footer

- **Wall-clock (this report build session):** 0.0 min.
- **Per-model cumulative compute (sum across 15 corruptions):** clip_vit_b16 7.0 min, convnext_tiny 3.4 min, efficientnet_b0 1.8 min, resnet50 2.9 min, swin_t 3.9 min, vgg16 3.5 min, vit_b_16 6.8 min.
- **Total compute (sum across model × corruption):** 29.3 min.
- **Seed:** 42 (set on random, numpy, torch).
- **Re-run:** `python scripts/generate_datasets.py --corruptions --severity 3` then `python scripts/run_corruptions_benchmark.py` then `python scripts/build_corruptions_report.py` then `python scripts/build_gradient_report_pdf.py --input results/corruptions/REPORT.md --output results/corruptions/REPORT.pdf`. Per-(model, corruption) JSONs are the resumption unit — delete one to force its recomputation.